Patent phrase matching

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DRAFT: scores verified through Kaggle API; required Submissions-tab screenshot is missing. Course-platform upload remains pending.

1. Validation design and baselines

I used DeBERTa-v3-small to predict similarity for an anchor-target pair in CPC context, with a regression head and mean squared error. The seed-42 GroupShuffleSplit holds out 20% of anchor groups. A random row split could place the same anchor in training and validation with different targets, making performance on unseen anchors look better than it is.

Run / changed factor	Validation r	Training min	Peak GB
section_baseline none (starter)	0.8202	3.43	2.73
minilm_no_training bi-encoder; different pretrained checkpoint; no task training	0.5814	0.00	-
context_none context only: none	0.8167	3.33	2.66
context_code context only: code	0.8192	3.31	2.71

The pretrained MiniLM bi-encoder, with no task training, obtained r=0.5814; the fine-tuned cross-encoder obtained r=0.8202 (difference +0.2387). MiniLM encodes each phrase separately and compares normalized mean-pooled embeddings. The cross-encoder attends jointly across both phrases. The checkpoint and architecture also change, so I cannot attribute the full gap to fine-tuning.

Training: maximum length 64, batch 32, three epochs, learning rate 2e-5, warmup ratio 0.1, weight decay 0.01. Timings cover Trainer.train(), including epoch evaluation; MiniLM has no training time. Peak memory is allocated CUDA memory, excluding other processes and some reserved memory.

2. Controlled context experiments

context_none: I changed only the context representation; delta r=-0.0035. The difference is below 0.005. Another seed would help check whether it persists.

context_code: I changed only the context representation; delta r=-0.0010. The difference is below 0.005. Another seed would help check whether it persists.

I selected section_baseline for the final run using validation r=0.8202. All variants use the same validation IDs and initial checkpoint. No leaderboard score was used for selection.

3. Section and score analysis

Section	N	Pearson r
A	943	0.7858
B	1926	0.8938
C	1177	0.7692
D	199	0.7892
E	373	0.8088
F	968	0.8475
G	1040	0.7062
H	933	0.8050

The lowest measured section is G (n=1040). Five of the ten largest errors are in G. The smallest group, D (199 pairs), is not the lowest. Coarse context may leave technical relations ambiguous, but these averages alone cannot explain the errors.

True score	N	Mean prediction
0.00	1522	0.119
0.25	2465	0.305
0.50	2475	0.492
0.75	832	0.661
1.00	265	1.025

Zero labels move inward and 0.75 is underpredicted, but the mean at 1 overshoots: compression is not uniform. The dominant 0.25/0.5 labels and squared-error regression can contribute to central tendency, but do not explain every group. These are raw predictions; only submission predictions are clipped to [0,1]. Pearson alone does not establish calibration.

4. Largest validation errors

I reviewed the ten largest absolute errors from the validation-selected model and grouped them below. A large prediction error alone is not enough to judge an expert label incorrect.

Anchor / target / context	True / predicted	Reviewed failure and evidence
punch face / face punch [B05]	0.00 / 0.917	Relation-sensitive lexical match: Reversing 'punch face' to 'face punch' can change a tool surface into an action. The high score is consistent with lexical confusion; its cause is not established. Label debatable: no clear issue
boom hydraulic cylinder / boom a hydraulic cylinder [E02]	0.00 / 0.830	Relation-sensitive lexical match: The target inserts 'a' into almost the same content words. The model scores 0.830 despite the zero label. Label debatable: possibly; expert context needed
imaging axis / imaging of axis [G02]	0.00 / 0.820	Relation-sensitive lexical match: An imaging axis names an axis; imaging of an axis names a process applied to it. The model largely treats them alike. Label debatable: no clear issue
imaging axis / image of axis [G02]	0.00 / 0.766	Relation-sensitive lexical match: An imaging axis and an image of an axis need not denote the same thing; adding 'of' changes the relation. Label debatable: no clear issue
end pins / wood pin [G10]	0.75 / -0.008	Technical or part-whole relation: The anchor names a pin component while the target adds a material/type. The model misses the annotated strong relation. Label debatable: possibly; expert context needed
breech mechanism / breech loading naval gun [B29]	0.75 / 0.005	Technical or part-whole relation: A breech mechanism is a component; the target names a breech-loading gun. Related granularity is missed, with prediction 0.005. Label debatable: no clear issue
average pore size / average cell size [B24]	0.75 / 0.040	Technical or part-whole relation: Both phrases measure structural size, but 'pore' versus 'cell' needs a technical sense. Prediction 0.040 misses the 0.75 label. Label debatable: possibly; expert context needed
return structure / return to structure [D01]	0.00 / 0.698	Relation-sensitive lexical match: 'Return structure' can name a structure; 'return to structure' expresses a direction/action. The high prediction misses this possible relation change. Label debatable: no clear issue
whose identity / customer id [G06]	0.75 / 0.066	Functional paraphrase missed: The phrases concern identity versus a customer identifier, but the wording and specificity differ. The model misses annotated relatedness. Label debatable: possibly; expert context needed
running tally / credit total [G08]	0.75 / 0.099	Functional paraphrase missed: A running tally and a credit total both concern accumulated quantities; little literal overlap accompanies a low prediction. Label debatable: no clear issue

The companion CSVs contain the full review notes and predictions. These categories describe ten selected failures; they do not show how common each error is across the validation set.

5. Late submissions and score comparison

Version / status	Public r	Private r
1 / complete	0.7922	0.8084

Version 1: public-minus-validation -0.0280; private-minus-validation -0.0118.

The final saved run independently reproduces validation r=0.820194478; its Trainer-reported runtime is 214.4694 seconds. For the recorded submission, both hidden scores are lower and the private result is closer to validation.

Different anchor mixes, CPC sections and partition sizes can contribute to these score gaps. Choosing settings on validation can also make that score optimistic. I kept the validation-selected configuration and training budget for the final notebook. It loads attached weights with internet off and writes the required id,score columns in submission.csv.

Every submission is listed above, including failures. Version URLs and screenshot evidence:

1: https://www.kaggle.com/code/utkrishtsuman/a24ariu0001-patent-final?scriptVersionId=367835971
Screenshot: MISSING
Notes: Submission 58176236; all attempts listed (one). Scores verified using official Kaggle API. Submissions-tab screenshot missing.

References and attribution

Course patent_phrase_starter.ipynb, download_weights.ipynb and project_brief.md, bk-anupam/generative_ai_shared (adapted starter code).

Kaggle: U.S. Patent Phrase to Phrase Matching, competition data and rules. https://www.kaggle.com/competitions/us-patent-phrase-to-phrase-matching

He et al. (2021). DeBERTaV3. arXiv:2111.09543. Model: microsoft/deberta-v3-small (MIT).

Reimers & Gurevych (2019). Sentence-BERT. arXiv:1908.10084. Model: sentence-transformers/all-MiniLM-L6-v2 (Apache-2.0).

Aslanyan & Wetherbee (2022). Patents Phrase to Phrase Semantic Matching Dataset. arXiv:2208.01171. Only competition train/test files used; no released test labels.